Cohen 2021 multiple myeloma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=5) fuses the patient-private myeloma clones: iLISI median 1.0→12.5 / 54 samples, one connected kNN component. Judged on the regenerated embedding across a θ×n_pcs sweep — samples intermix on one continuous, low-cycling body (no proliferative lobe to scope). θ=5 keeps the manifold axis-bearing; θ=10 over-smooths to an axis-less blob.

Gate 2 — module geography → GO (non-branching)

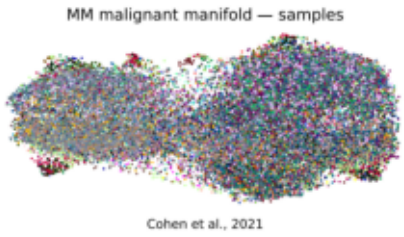
Signatures resolve onto a single (Gast/pSom)→Crest axis: smoothed Gast↔Crest r=−0.73, pSom↔Crest r=−0.65, Gast↔pSom r=+0.28 (overlap, not two competing arms). Crest is an inverse terminal gradient (Crest-high decile pseudotime 0.85 vs Gast 0.31 / pSom 0.15). Principal curve is fold-free (2 tips / 0 forks, self-approach 2.8). No Crest-on-Gast-arm inversion to flag.

Gate 3 — cross-repression → NA

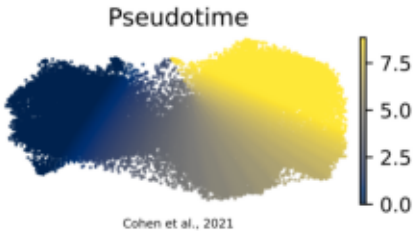
Non-branching topology — no bifurcation and no Gast/pSom hub, so the branching-only cross-repression test does not apply. Correlation & modules notebooks are correctly not produced.

STEP FIGURES

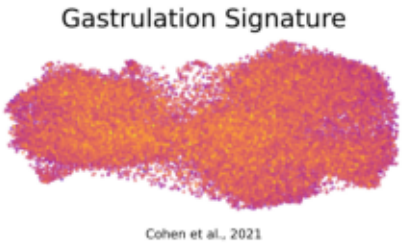
Manifold — samples



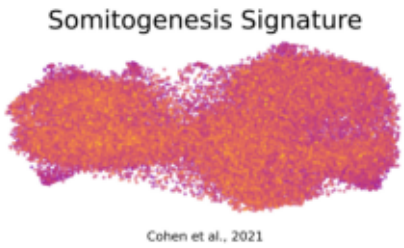
Pseudotime



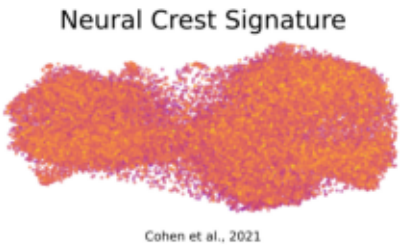
Gastrulation



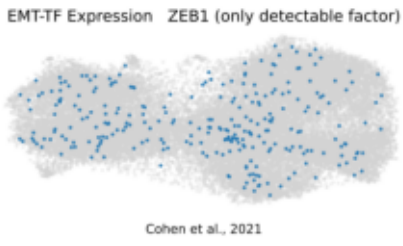
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Multiple myeloma (Cohen et al., 2021), MARS-seq, 32,894 malignant plasma cells over 54 samples. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces as the non-branching case — a single gastrulation/somitogenesis → neural-crest developmental axis; no hub, no bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TF ZEB1 is the only factor above the 1% detection floor (terminally-differentiated plasma cells).

Parameter choices

Min-cell threshold	≥100 floor (no dominant cliff); 54/73 samples kept	**Integration**	Harmony on sample θ=5; n_pcs=30, n_neighbors=30, spread=1.3
Islands removed	253 cells / 14 detached specks (largest component)	**Geometry**	Non-branching; scFates curve N=20, λ=0.3, μ=100 (2 tips / 0 forks)
Root node	curve tip 2 = Gast/pSom-high, Crest-low (Crest-high at terminus)	**Module means (raw)**	Gast −0.022 · pSom +0.014 · Crest +0.001 (Linear single axis)
EMT-TFs shown	ZEB1 (3.2%) — only EMT-TF above the 1% floor; all others <0.3%		